

CLARESHOLM, AB, Canada

WMO: 712340

Lat: 50.0036N

Lon: 113.6406W

Elev: 3310

StdP: 13.02

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-14.5	-27.6	1.3	-20.6	-20.8	2.0	-14.2	39.3	44.9	31.9	41.4	3.3	0	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.2	87.2	60.4	83.6	59.6	80.1	58.7	65.0	78.5	62.9	77.0	61.1	75.4	10.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	88.0	70.9	57.7	80.4	68.6	55.5	74.3	65.9	31.8	78.7	30.1	76.7	28.7	75.4	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.8	23.5	20.3	DB	-30.8	93.4	6.3	3.4	-35.4	95.9	-39.1	97.8	-42.6	99.7	-47.2	102.2
			WB	-31.0	69.6	6.3	4.5	-35.5	72.9	-39.2	75.6	-42.7	78.1	-47.3	81.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	20.5	22.1	30.1	40.5	50.2	57.4	63.8	62.3	53.9	42.3	30.0	21.0	(6)
(7)		DBStd	19.33	16.88	15.43	14.26	9.11	7.88	5.75	5.39	5.89	8.04	9.63	14.01	15.84	(7)
(8)		HDD50	4545	914	783	619	301	95	7	1	2	51	269	603	900	(8)
(9)		HDD65	8761	1379	1203	1082	736	461	235	86	121	340	704	1049	1365	(9)
(10)		CDD50	1361	0	0	2	16	101	230	428	383	167	30	4	0	(10)
(11)		CDD65	102	0	0	0	0	2	8	49	37	6	0	0	0	(11)
(12)		CDH74	2556	0	0	0	7	116	204	979	925	316	9	0	0	(12)
(13)		CDH80	767	0	0	0	0	18	38	318	308	85	0	0	0	(13)
(14)	Wind	WSAvg	7.9	8.3	7.6	8.5	8.5	8.3	7.9	7.2	6.7	7.1	8.1	8.7	8.1	(14)
(15)	Precipitation	PrecAvg	16.5	0.4	0.5	0.8	1.1	2.5	3.7	2.0	1.8	1.7	0.8	0.7	0.6	(15)
(16)		PrecMax	29.6	0.9	1.4	2.5	2.4	5.1	14.5	5.7	4.0	5.3	3.0	2.0	1.3	(16)
(17)		PrecMin	7.0	0.0	0.0	0.0	0.2	0.0	0.5	0.0	0.0	0.2	0.1	0.1	0.0	(17)
(18)		PrecStd	4.8	0.2	0.3	0.6	0.7	1.4	2.8	1.5	1.0	1.1	0.6	0.5	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.0	63.9	74.0	82.7	85.1	91.9	91.7	87.2	74.9	64.6	54.7	(19)
MCWB			41.7	42.2	45.5	50.5	56.4	62.8	62.6	60.8	57.7	52.5	47.7	42.2	(20)	
2%		DB	49.9	51.4	58.4	67.8	77.2	79.5	87.5	87.2	82.2	68.7	57.8	49.1	(21)	
		MCWB	39.4	39.5	43.5	48.2	54.4	58.9	61.6	59.6	56.2	50.3	44.5	39.0	(22)	
5%		DB	46.4	46.8	54.0	62.5	72.2	75.6	83.4	83.4	77.3	63.1	53.0	45.5	(23)	
		MCWB	37.6	36.8	41.2	45.6	52.4	57.4	61.3	59.1	55.3	48.0	42.1	36.8	(24)	
10%		DB	43.1	42.9	49.6	57.4	67.5	72.1	79.4	79.3	72.0	58.3	48.7	42.1	(25)	
		MCWB	35.4	34.4	38.8	43.1	50.5	55.5	60.5	58.5	53.5	45.3	39.0	34.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	43.2	47.4	52.0	58.4	66.7	68.8	65.6	60.9	54.8	49.4	43.4	(27)
MADB			52.5	54.0	61.2	71.0	76.1	78.3	82.4	80.0	79.7	70.6	61.2	52.8	(28)	
2%		WB	40.2	40.2	44.3	49.1	55.7	62.0	66.0	63.3	58.2	51.3	45.5	39.9	(29)	
		MADB	48.7	50.7	56.4	65.4	73.4	74.1	79.0	78.9	76.6	66.2	56.8	48.0	(30)	
5%		WB	37.9	37.3	41.8	46.5	53.7	59.4	64.1	61.7	56.2	48.4	42.4	37.1	(31)	
		MADB	45.6	46.4	53.4	61.3	69.5	71.9	77.7	77.5	73.9	61.7	52.4	45.0	(32)	
10%		WB	35.7	34.7	39.1	43.8	51.5	57.2	62.1	60.0	54.1	45.8	39.5	34.5	(33)	
		MADB	43.0	42.4	49.1	56.3	65.7	68.8	75.6	75.2	69.9	57.1	48.3	41.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.2	22.7	22.4	23.7	24.5	23.5	27.2	28.6	27.4	24.3	21.5	21.8	(35)
MCDBR			23.4	26.5	28.9	34.0	34.3	31.0	34.9	37.5	38.8	31.9	25.9	22.6	(36)	
MCWBR			15.5	16.5	16.6	17.7	15.7	15.1	15.1	15.2	17.1	17.0	15.6	15.0	(37)	
5% WB		MCDBR	22.9	26.7	28.6	31.5	30.5	27.7	29.9	31.2	34.5	30.2	25.8	22.8	(38)	
		MCWBR	15.8	17.4	17.1	16.8	14.3	15.2	15.1	14.1	16.0	16.7	15.8	15.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.262	0.285	0.328	0.337	0.333	0.337	0.352	0.314	0.283	0.260	0.235	(40)	
(41)		taud	2.440	2.473	2.466	2.355	2.391	2.445	2.448	2.393	2.478	2.535	2.496	2.453	(41)	
(42)		Ebn at Noon	259	287	296	291	292	293	290	280	282	274	256	253	(42)	
(43)		Edh at Noon	19	23	28	36	36	34	34	34	28	22	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	384	699	1096	1520	1808	1990	2196	1758	1244	770	421	298	(44)	
(45)		RadStd	31	58	88	105	143	141	120	113	112	82	36	27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page